TOP 50 BIGQUERY CONCEPT QUESTIONS

1 What is Google BigQuery?

BigQuery is Google Cloud's **fully managed, serverless data warehouse**. You run fast **SQL analytics** on datasets from gigabytes to **petabytes** without managing any servers.

Key Benefit

✓ Scalable, fast, serverless and pay-as-you-go analytics.

2 Why is BigQuery called serverless?

You never create, size or patch servers. Google allocates compute automatically for **every query** and scales it up or down. You only write SQL and pay for what you use.

Remember

★ No clusters. No VMs. No hardware tuning. **Just SQL.**

3 What is the difference between BigQuery (OLAP) and a normal database (OLTP)?

Aspect	OLTP (Cloud SQL, Spanner)	OLAP (BigQuery)
Purpose	Run apps and transactions	Analyze and report on data
Workload	Many small reads / writes of single rows	Few huge scans, aggregations, joins
Storage	Row-oriented	**Column-oriented**
Best for	Apps, orders, payments	BI, dashboards, ML on big data

Think: Transactions go to Cloud SQL / Spanner. Analytics go to BigQuery.

4 What are the main components of BigQuery architecture?

Dremel
Query engine (execution tree)

Colossus
Distributed file system for storage

Capacitor
Columnar storage format

Jupiter
Fast datacenter network

Borg
Cluster manager for resources

Think: Compute (Dremel) + Storage (Colossus) + Network (Jupiter) = BigQuery.

5 What is separation of storage and compute?

Data is stored in **Colossus** and queries run on separate **Dremel** compute, linked by the Jupiter network. Each side scales on its own.

Key Point

✓ Store huge data cheaply; add compute only while queries run.

Storage
Colossus
→
Jupiter network
petabit speed
→
Compute
Dremel slots

6 What is columnar storage and why is it fast?

	Row-oriented	Column-oriented (BigQuery)
Layout	Whole row stored together	Each column stored together
Query on 3 of 50 columns	Reads everything	**Reads only 3 columns**
Compression	Weaker	**Very high** (same-type data)

Remember

★ **SELECT *** reads every column, so it is slow and expensive. Name only the columns you need.

7 What is a slot?

A **slot** is BigQuery's unit of compute (virtual CPU, memory, I/O) used to run query stages. On-demand projects share a pool (by default about **2,000 slots**); capacity customers buy or autoscale slots.

Key Point

✓ More slots = more parallel work = faster complex queries.

8 What is a Project in BigQuery?

A **Project** is the top container for your resources. It holds datasets, handles **billing, quotas and permissions**, and runs your query jobs.

Remember

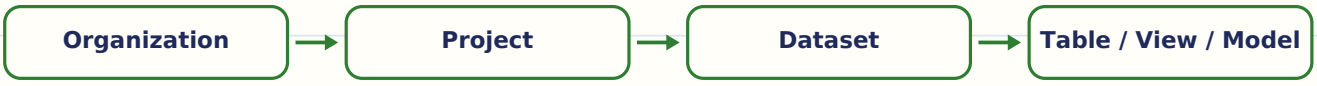
★ All BigQuery resources must belong to a Project.

9 What is a Dataset?

A **Dataset** is a container inside a project that holds tables, views, routines and models. It has a **location**, access controls and optional default expiration.

Key Path

✓ Project > Dataset > Table. Full name: `project.dataset.table`



10 What are the different table types?

Type	Meaning
Native table	Data stored inside BigQuery
External table	Data stays in Cloud Storage / Drive / Bigtable
View	Saved SQL query, no data stored
Materialized view	Pre-computed, auto-refreshed result
Snapshot / Clone	Read-only / writable lightweight copy
Temporary table	Query results, kept about 24 hours

11 What are jobs in BigQuery?

Load
files into a table

Query
run SQL

Copy
table to table

Extract
table to Cloud Storage

Note: Every action runs as an **asynchronous job** with a job id. Load, copy and extract jobs are free on shared slots.

12 Region vs multi-region location?

Location	Meaning
Region	One place, e.g. asia-south1 (Mumbai)
Multi-region	Large area: **US** or **EU**

Remember
★ Dataset location **cannot be changed**. A query can only join datasets in the **same location**.

13 What data types does BigQuery support?

Numeric
INT64, FLOAT64, NUMERIC, BIGNUMERIC

Text
STRING, BYTES

Date / Time
DATE, TIME, DATETIME, TIMESTAMP

Complex
ARRAY, STRUCT, JSON, GEOGRAPHY

Tip: Use **NUMERIC** for money (exact). TIMESTAMP is a UTC moment; DATETIME has no time zone.

14 What are STRUCT and ARRAY (nested and repeated data)?

STRUCT = a record with named fields. **ARRAY** = a list of values. An **ARRAY of STRUCT** stores one-to-many data inside one row, avoiding joins.

Remember
★ Denormalized, nested data is the **BigQuery way**.

NESTED SCHEMA

```
CREATE TABLE my_ds.orders (order_id INT64, customer STRING,
  items ARRAY<STRUCT<name STRING, qty INT64, price NUMERIC>>);
```

15 What does UNNEST do?

UNNEST flattens an array into rows - one row per element - so you can filter and join array items.

Key Point
✓ Think of it as "explode the array".

EXAMPLE

```
SELECT o.order_id, i.name, i.qty
FROM my_ds.orders AS o, UNNEST(o.items) AS i
WHERE i.qty > 1;
```

16 GoogleSQL vs Legacy SQL?

	GoogleSQL (Standard SQL)	Legacy SQL
Status	**Default, recommended**	Old dialect, avoid for new work
Standard	ANSI SQL 2011 compliant	BigQuery-specific syntax
Features	Window functions, arrays, structs, scripting, MERGE	Limited

17 What is a CTE (WITH clause)?

A **Common Table Expression** names a sub-query so the main query is easier to read and reuse.

Key Point

✓ WITH name AS (...) SELECT ... FROM name

EXAMPLE

```
WITH monthly AS (
  SELECT DATE_TRUNC(order_date, MONTH) AS m, SUM(amount) AS total
  FROM my_ds.sales GROUP BY m)
SELECT * FROM monthly ORDER BY m;
```

18 What are window functions?

Calculate across related rows **without collapsing them** (unlike GROUP BY). Syntax: `function() OVER (PARTITION BY ... ORDER BY ...)`

Function	Use
ROW_NUMBER / RANK / DENSE_RANK	Number or rank rows within a group
LAG / LEAD	Previous / next row value
SUM / AVG OVER	Running totals, moving averages
NTILE(n)	Split rows into n buckets

19 What is QUALIFY?

```
-- top 3 orders per region
SELECT * FROM my_ds.sales
QUALIFY ROW_NUMBER() OVER (PARTITION BY region ORDER BY amount DESC) <= 3;
```

★ **Golden Rule:** QUALIFY is to window functions what HAVING is to GROUP BY - no sub-query needed!

20 What are the types of JOINs?

Join	Returns
INNER	Only rows that match in both tables
LEFT	All left rows + matches from right (NULL if none)
RIGHT	All right rows + matches from left
FULL OUTER	All rows from both sides
CROSS	Every combination (use with care)

21 View vs Materialized View?

	View	Materialized View
Stores data?	No, saved query	**Yes**, pre-computed result
Refresh	Runs fresh every time	**Auto-refreshed** (about 30 min default)
Cost / speed	Pays for base table scan	Faster and cheaper (incremental)
SQL allowed	Any query	Limited (aggregations, simple joins)

22 What is an Authorized View?

A view that can read a dataset the user **cannot access directly**. Share a filtered or masked slice of data without exposing the base table.

Key Point

✓ Great for sharing safely across teams.

User (no table access) → **Authorized View filters columns / rows** → **Base table stays private**

23 What is partitioning? What types exist?

Partitioning splits a table into **segments**. Queries that filter on the partition column read only matching partitions - this saves cost and time.

Time-unit column

DATE / TIMESTAMP, by hour / day / month / year

Ingestion time

_PARTITIONTIME load time

Integer range

ranges of an INT64 column

Limit: Up to **10,000 partitions** per table and **one** partition column.

24 What is clustering?

Clustering sorts data **inside** a table or partition by up to **4 columns**, so BigQuery skips blocks that do not match your filters. Re-clustering is automatic and free.

Remember

★ **Column order matters**: the first cluster column sorts first.

25 Partitioning vs Clustering?

	Partitioning	Clustering
Columns	One	Up to 4
Best on	Low-cardinality (date)	High-cardinality (id, region)
Cost estimate	**Exact** before running	Upper bound only
Together?	Yes - partition by date, cluster by id / region	

★ **Golden Rule:** Partition on **date**. Cluster on what you **filter by most**.

26 What is partition pruning and require_partition_filter?

Pruning: BigQuery skips partitions your WHERE clause excludes. **require_partition_filter** forces every query to filter on the partition column - no accidental full scans.

Key Benefit

✓ Stops costly full-table scans.

DDL

```
CREATE TABLE my_ds.sales (order_id INT64, order_date DATE, region STRING)
PARTITION BY order_date CLUSTER BY region
OPTIONS (require_partition_filter = TRUE, partition_expiration_days = 365);
```

27 Sharded tables vs partitioned tables?

	Sharded (events_20260101)	Partitioned (one table)
Management	Many tables to secure and maintain	One table
Querying	Wildcard + _TABLE_SUFFIX	Normal WHERE on partition column
Performance	Slower (metadata per table)	**Faster**
Recommended	No	**Yes**

28 How can you load data into BigQuery?

Method	Best for
Batch load (free)	Files from Cloud Storage - daily / hourly
Streaming / Storage Write API	Real-time events
Data Transfer Service	Google Ads, GA4, YouTube, S3 ...
Dataflow / Dataproc	ETL pipelines
Pub/Sub subscription	Messages straight into a table
Datastream	CDC from MySQL / Postgres / Oracle

29 Batch load vs streaming?

	Batch load	Streaming (Storage Write API)
Cost	**Free**	Charged (free monthly allowance)
Latency	Minutes - hours	Seconds
Delivery	Atomic job	Exactly-once option
Use for	Scheduled files	Live dashboards, events

Remember

★ The **Storage Write API** is the recommended way to stream; legacy insertAll is older.

30 Which file formats are supported? Which is best?

Format	Note
Avro	Self-describing, fast, preferred
Parquet / ORC	Columnar, compressed
CSV	Needs schema / auto-detect
JSON	Newline-delimited only

Key Point

✓ Prefer **Avro** or **Parquet**: schema is inside and data is compressed.

31 What is an external table and a BigLake table?

An **external table** reads files in place (Cloud Storage, Drive, Bigtable) - nothing is loaded. A **BigLake table** adds fine-grained security and better performance for lake data.

Remember

★ Native tables are fastest. Use external tables for staging or rare queries.

32 What is a federated query?

Run SQL on **Cloud SQL, Spanner or AlloyDB** from BigQuery using EXTERNAL_QUERY - without copying the data.

Key Point

✓ Join live operational data with warehouse tables.

```
SELECT * FROM EXTERNAL_QUERY("my-project.us.my_connection",
  "SELECT id, name FROM customers WHERE active = 1");
```

33 What are Data Transfer Service and scheduled queries?

Data Transfer Service

Scheduled, managed imports from **Google Ads, GA4, YouTube, Amazon S3, Teradata** and more.

Scheduled queries

Run SQL **hourly / daily** and write results to a table - perfect for refreshing report tables.

34 What are BigQuery Omni and Analytics Hub?

Service	What it does
BigQuery Omni	Query data in **AWS S3 / Azure Blob** without moving it
Analytics Hub	Share and subscribe to datasets through exchanges and listings (linked datasets)

35 How is BigQuery priced?

	On-demand (default)	Capacity / Editions
You pay	About **$6.25 per TiB** scanned (US)	**Slot-hours** (reserved / autoscaled)
Free tier	**First 1 TiB / month free**	None
Predictable?	Varies with data scanned	**Yes**
Best for	Light or spiky use	Heavy, steady workloads

Note: Approximate US list prices - confirm on the Google Cloud pricing page.

36 What are BigQuery Editions?

Edition	For
Standard	Basic workloads, autoscaling, no commitments
Enterprise	Production analytics: governance, security, commitments
Enterprise Plus	Mission-critical: highest availability, compliance

37 Active vs long-term storage?



Key Benefit

✓ Automatic, same performance and durability. Each **partition** ages separately. First **10 GiB** free each month.

38 Logical vs physical storage billing?

	Logical	Physical
Counts	Uncompressed bytes	**Compressed** bytes
Cheaper when	Data barely compresses	Data compresses well
Also bills	-	Time-travel and fail-safe bytes

Choice: Set per dataset. Compare both before switching.

39 Which BigQuery operations are free?

Free

- Batch **load**, **export**, **copy**
- Metadata operations, dry runs
- **Cached** query results
- First **1 TiB** queried and **10 GiB** stored per month

Paid

- Queries (per TiB or slots)
- Streaming / Storage Write API (after free allowance)
- BI Engine, BigQuery ML, Omni
- Storage above the free 10 GiB

40 How can you reduce BigQuery cost?

#	Tip
1	Select only needed columns - avoid **SELECT ***
2	Filter on the **partition** column; cluster by filter columns
3	Use **materialized views** and the query cache
4	Preview with **dry run**; set **maximum bytes billed**
5	Use **require_partition_filter**, expiration, budgets and alerts
6	Use approximate functions (`APPROX_COUNT_DISTINCT`)

41 Does LIMIT reduce cost? What is a dry run?

No. On a normal table BigQuery still scans every selected column. A **dry run** validates the query and shows bytes it would process - it costs **nothing**.

Remember

★ Cost depends on **bytes scanned**, not rows returned.

```
bq query --dry_run --use_legacy_sql=false "SELECT region FROM my_ds.sales"
```

42 What is the query cache?

BigQuery stores results of a query for about **24 hours**. Running the same query again is **free** and instant.

Conditions

★ Not used if tables changed or the query uses **RAND / CURRENT_TIMESTAMP**.

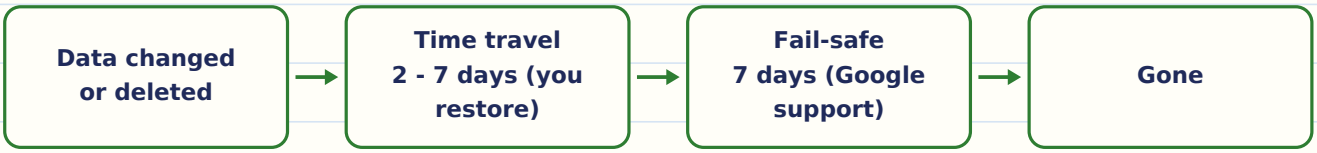
43 What is BI Engine?

An **in-memory acceleration** layer that gives sub-second responses to dashboards in **Looker Studio, Looker, Sheets** and other BI tools. You reserve GB of memory.

Key Benefit

✓ Fast dashboards without changing queries.

44 What are time travel and fail-safe?




```
SELECT * FROM my_ds.orders
FOR SYSTEM_TIME AS OF TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 1 HOUR);
```

45 Snapshot vs Clone?

	Table snapshot	Table clone
Mode	**Read-only**	**Read / write**
Use	Backup, point-in-time copy	Dev, testing, experiments
Storage cost	Only changed data	Only changed data

Key Point: Both are copy-on-write: instant and almost free until data changes.

46 What are the main IAM roles in BigQuery?

Role	Can do
Admin	Everything
Data Owner	Full control of datasets / tables, grant access
Data Editor	Read and write table data
Data Viewer	Read data and metadata
Job User / User	Run query jobs in the project

★ **Golden Rule:** To query a table you need **Data Viewer** on the data AND **Job User** on a project.

47 Row-level vs column-level security and data masking?

Feature	How it works
Row-level	Row access policies filter rows per user / group
Column-level	Policy tags (Dataplex / Data Catalog) restrict columns
Data masking	Shows hashed / nulled / partial values to some users

```
CREATE ROW ACCESS POLICY south_only ON my_ds.sales
GRANT TO ("group:south-team@company.com") FILTER USING (region = "South");
```

48 How is data encrypted and audited?

Area	Feature
At rest	**Always encrypted**; optional **CMEK** (Cloud KMS keys)
In transit	TLS
Audit	**Cloud Audit Logs**: who ran which query
Perimeter	**VPC Service Controls**
Governance	Dataplex / Data Catalog, Sensitive Data Protection (DLP)

49 What is BigQuery ML?

Create, train and use ML models **with only SQL**, where the data lives. Models: linear / logistic regression, k-means, boosted trees, DNN, ARIMA_PLUS, matrix factorization, imported and remote (Gemini).

Key Benefit

✓ No data export, no separate ML infrastructure.

TRAIN + PREDICT

```
CREATE OR REPLACE MODEL my_ds.churn OPTIONS (model_type = "logistic_reg", input_label_cols = ["churned"])
AS SELECT tenure, monthly_charges, churned FROM my_ds.customers;
SELECT * FROM ML.PREDICT(MODEL my_ds.churn, (SELECT * FROM my_ds.new_customers));
```

50 BigQuery vs Snowflake vs Redshift vs Synapse?

	BigQuery	Snowflake	Redshift	Synapse / Fabric
Cloud	Google	AWS, Azure, GCP	AWS	Azure
Model	Fully serverless	Virtual warehouses	Clusters / Serverless	Pools / Fabric capacity
Strength	Zero ops, built-in ML, streaming	Multi-cloud sharing	AWS ecosystem	Microsoft + Power BI

★ **Golden Rule:** Scan less data = pay less and run faster. Read **fewer columns** and **fewer partitions**!